

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT3 — VISUALIZATION CRITIQUE & REDESIGN

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Critique & Redesign
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Evaluate flawed or misleading data visualizations, critique their specific shortcomings, and justify an improved redesign. (BL5)</i>		
Student Name:	PRIYANKA G	Reg. No.:	192524124
Section / Batch:		Date:	18.08.2026

Code of Conduct

I Priyanka.G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond simply naming a flaw: explain WHY it is misleading and what a reader could wrongly conclude from it.
- Your redesign must be a specific, justified chart proposal — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION CRITIQUE & REDESIGN

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5 Color Perception Pie Chart of Near Equal Categories (5 Marks)

Scenario: A pie chart uses red, orange, and yellow for three nearly equal sized categories, making the slices hard to visually tell apart at a glance.

Principle Violated	Color perception / hue discriminability in categorical encoding
Flaw Type	Analogous (adjacent hue) color palette used for unordered categories
Fix Type	Distinct qualitative palette + direct data labels

a) Critique this color choice.

Critique:

This chart violates the principle that colours used to distinguish unordered categories should be as perceptually distinct as possible. Red, orange, and yellow are analogous hues they sit right next to each other on the colour wheel and share similar lightness and saturation so the human eye has to work far harder to tell them apart than it would with hues spread further apart (e.g., blue, red, green). This problem is compounded here because the three categories are also nearly equal in size, so there is no area cue to compensate for the weak colour cue.

A reader glancing at this chart could easily misjudge the boundary between two adjacent slices, or fail to register that there are three distinct categories at all rather than two, and could wrongly conclude that one of the near equal categories is noticeably larger or smaller than the others simply because a misread colour boundary shifted where they perceived one slice ending and the next beginning.

In a real reporting context, this matters beyond aesthetics: if this pie chart were, say, comparing three product lines' market share to decide where to focus marketing spend, a viewer who misjudged the near equal slices as meaningfully unequal could recommend investing further in a product line that is not actually outperforming the others a decision built on a colour perception error rather than on the real underlying data.

192524124 Problem 5 - Flawed Pie Chart

Similar colors make nearly equal slices difficult to distinguish



Figure 5a: Flawed Pie Chart Red/Orange/Yellow, Difficult to Distinguish

b) Redesign: propose a better color scheme and explain your reasoning.

Redesign:

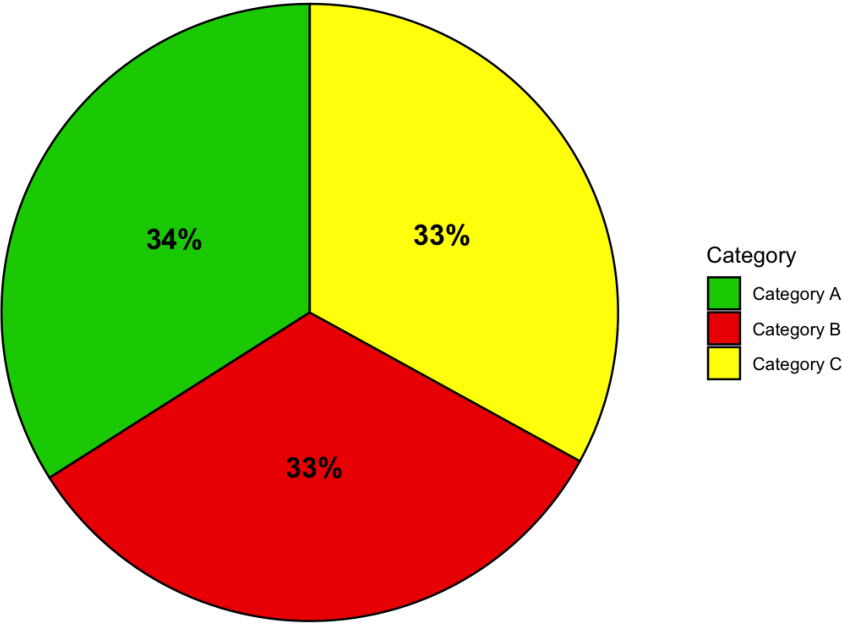
The redesigned chart replaces the analogous red orange yellow scheme with hues spread further apart around the colour wheel (green, red, and yellow), which are far easier to tell apart at a glance since they no longer share the same narrow hue family. Equally important, each slice is now directly labelled with its exact percentage (34%, 33%, 33%), so correctly reading each category's share no longer depends on colour discrimination at all – even a reader who struggled to distinguish the colours could still read the exact values directly off the chart.

This combination – distinct hues plus direct data labels – is the more robust fix specifically because it does not rely on any one perceptual channel working perfectly; the colour makes each slice locatable at a glance, while the label guarantees the precise value is never in doubt, which is why labelling is included as a core part of the redesign rather than being treated as optional.

More generally, this case illustrates a broader principle in colour perception for categorical data: hues chosen for unordered categories should be spread as far apart as possible in perceptual colour space (not merely 'different names of colour'), since human colour discrimination is far weaker for hues that are close together than for hues on opposite sides of the wheel – a rule that applies to any chart type using colour to separate categories, not pie charts alone.

192524124 Problem 5 - Redesigned Pie Chart

Distinct colors improve category identification



Interpretation: Clearly separated colors make the three nearly equal categories easier to distinguish.

Figure 5b: Redesigned Pie Chart Distinct Colours with Percentage Labels

Problem 6 Hidden Denominator 100% Stacked Bar of Department Gender Data (5 Marks)

Scenario: A report converts department gender data into a 100% stacked bar chart without showing the total headcount of each department.

Principle Violated	Percentage encodings must retain their base (denominator)
Flaw Type	Normalization to 100% removes absolute scale information
Fix Type	Plot actual counts (stacked by absolute headcount) instead of normalized percentages

a) Critique what critical information is hidden by this design.

Critique:

A 100% stacked bar chart normalizes every bar to the same total height by construction, regardless of how many people are actually in each group. This means the chart necessarily discards the one piece of information needed to judge real world scale each department's actual headcount in exchange for showing only the internal proportion (percentage) within each bar.

A reader looking at this flawed chart sees two bars of identical height (both 100%) and has no visual basis to suspect the departments might be very different sizes. They could easily and reasonably conclude that Science and Commerce have roughly comparable numbers of students or staff, when in fact as the redesign reveals one department could have twice the total headcount of the other; the percentage only format actively conceals exactly that difference.

This has real consequences if the chart is used to guide policy: a reader might conclude both departments need identical support for gender balance initiatives simply because the visualized proportions look comparably split, when the department with the larger headcount (Science, with 100 students) may need proportionally more resources allocated to actually shift its composition than the smaller department (Commerce, with 50) a distinction the 100% format makes impossible to see.

192524124 Problem 6 - Flawed 100% Stacked Bar Chart

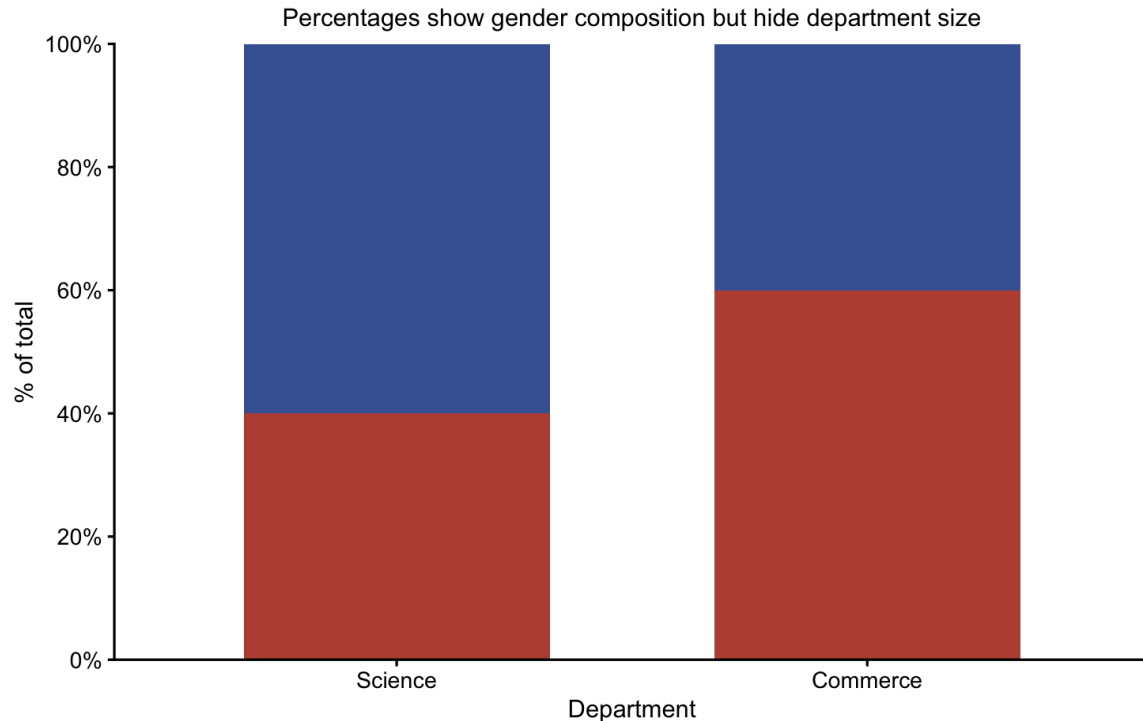


Figure 6a: Flawed 100% Stacked Bar Chart Department Sizes Hidden

b) Redesign: propose how to present the data so the hidden information becomes visible.

Redesign:

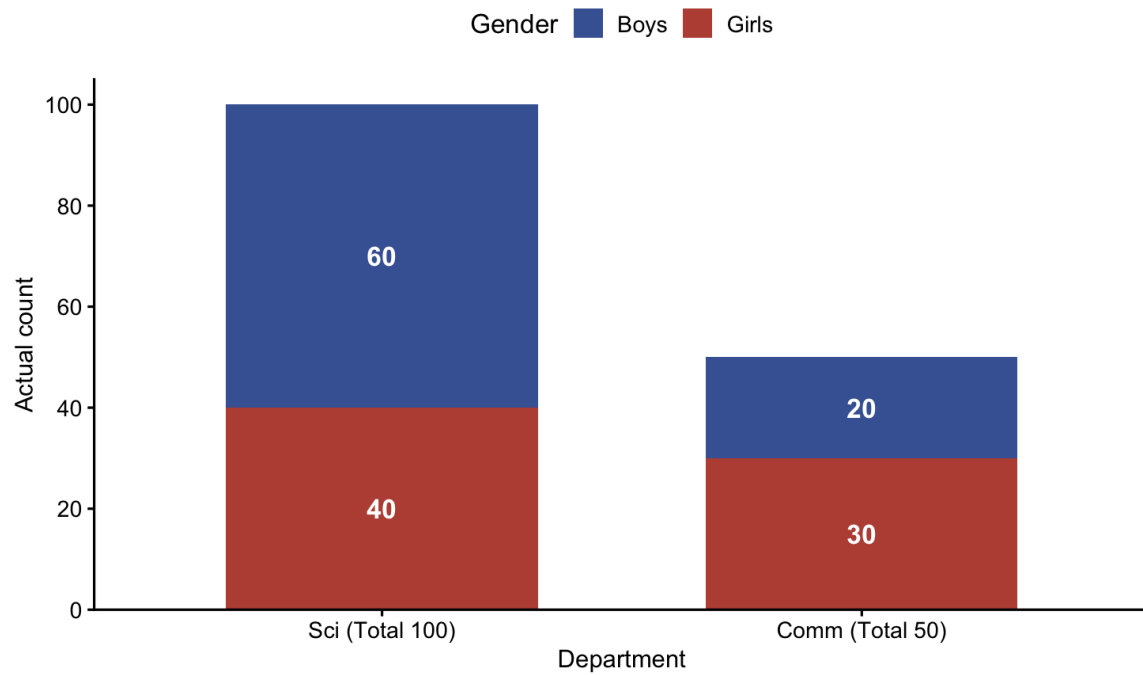
Instead of normalizing each bar to 100%, the redesigned chart plots actual counts: each bar's total height now equals that department's real headcount (Science = 100, Commerce = 50), and each segment within the bar shows the actual number of students of that gender (Science: 60 Boys / 40 Girls; Commerce: 20 Boys / 30 Girls), with the totals labelled directly on the axis categories.

This restores the department size comparison the original chart destroyed – a reader can now see at a glance that Science's bar is twice as tall as Commerce's, meaning Science has twice the total headcount – while still preserving the internal gender composition story the original 100% chart was trying to tell, since the relative segment heights within each bar are unchanged. No information is lost in this redesign; the missing denominator is simply added back.

As a general rule, any chart that reports a percentage or proportion should either display the underlying total alongside it (as done here) or, at minimum, state it clearly in a caption or label, since normalized encodings are only safe to compare across groups once the reader knows those groups are similar in scale – otherwise, as in the original chart, normalization can silently equate groups that are nothing alike in size.

192524124 Problem 6 - Redesigned Stacked Bar Chart

Actual department headcounts and gender composition are visible



headcount. Department A has 100 employees, Department B has 100 employees, and Department C has 200 employees.

Figure 6b: Redesigned Stacked Bar Chart Actual Headcounts Visible

Problem 12 Axis Scaling Density Plot of Household Income (5 Marks)

Scenario: A density plot of household income is shown with a tall spike near zero and a long, flat, uninformative tail, using a linear x axis.

Principle Violated	Axis scale must match the data's underlying (multiplicative) structure
Flaw Type	Linear x axis compresses a long tailed, multiplicative distribution into a single spike
Fix Type	Log10 transformation of the x axis

a) Critique why this chart fails to show the data's real structure.

Critique:

This chart violates the principle that an axis scale should be chosen to match how the underlying data actually varies. Household income is fundamentally multiplicative in nature incomes commonly differ by factors ($2\times$, $5\times$, $10\times$) rather than by fixed additive amounts and a small number of very high income households can be many multiples of a typical household's income. A linear x axis treats a ₹10,000 difference the same everywhere on the scale, whether it occurs between two low income households or is a rounding error next to a high income household, so the handful of extreme values stretch the axis out to values that squeeze the vast majority of ordinary households into one narrow bar near zero.

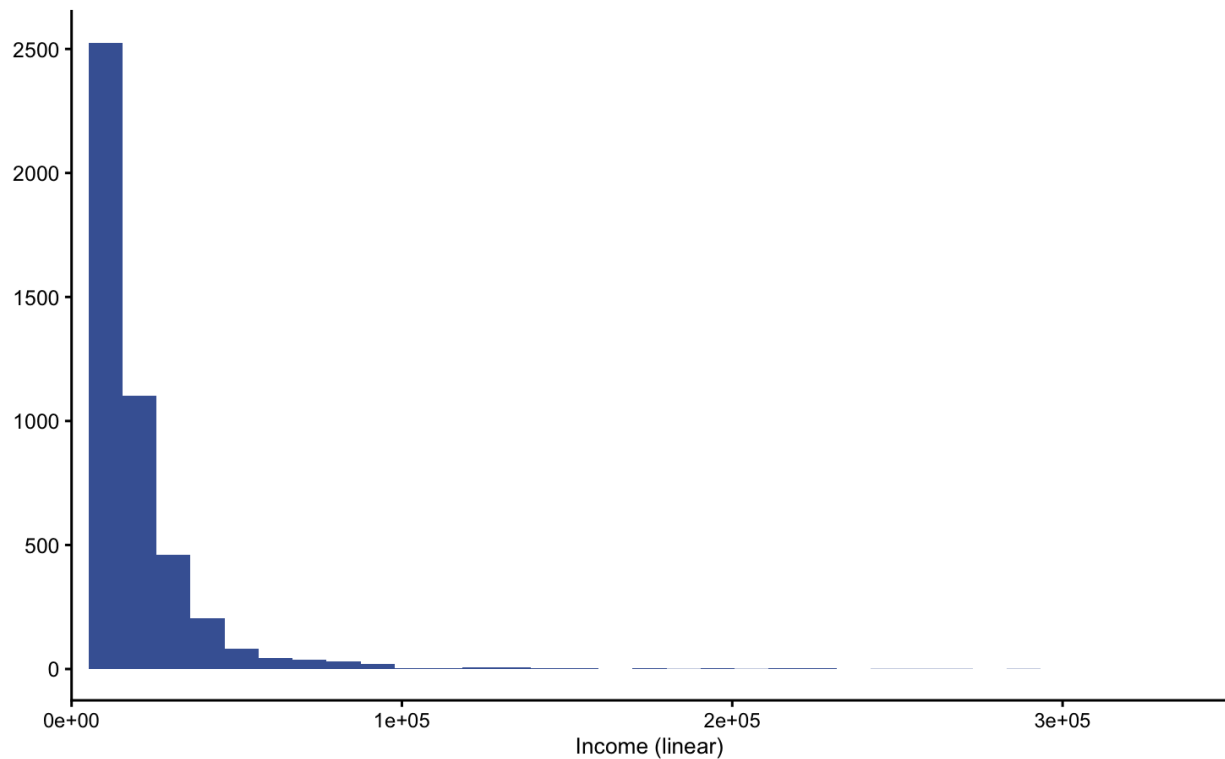
A reader looking at this flawed chart would see one tall spike and a long, almost invisible flat tail, and could easily conclude that nearly every household has essentially the same, very low income, with little worth examining beyond that single spike. In reality, there is meaningful variation across the bulk of the distribution it is simply compressed into invisibility by the choice of a linear scale, not because that variation doesn't exist.

This is a specific instance of a general axis scaling principle: whenever a variable's natural variation is multiplicative rather than additive commonly true of income, population, price, and many other real world quantities a linear axis will always compress the bulk of ordinary values against one edge in the presence of even a few extreme outliers, regardless of how many observations are involved.

If this chart were used to inform a policy decision for example, setting an income threshold for a subsidy programme a policymaker misled by the single spike could assume there is essentially one 'typical' household income with almost no variation worth accounting for, and set a single flat threshold accordingly, when the real distribution (visible only after the log transform) may call for a more nuanced, tiered approach.

192524124 Problem 12 - Flawed Income Distribution

Linear scale compresses most observations near zero



Interpretation: The long right tail dominates the linear scale, making the main concentration of household incomes difficult to examine.

Figure 12a: Flawed Income Density Plot Linear Scale Hides Structure

b) Redesign: propose a fix and explain why it would work.

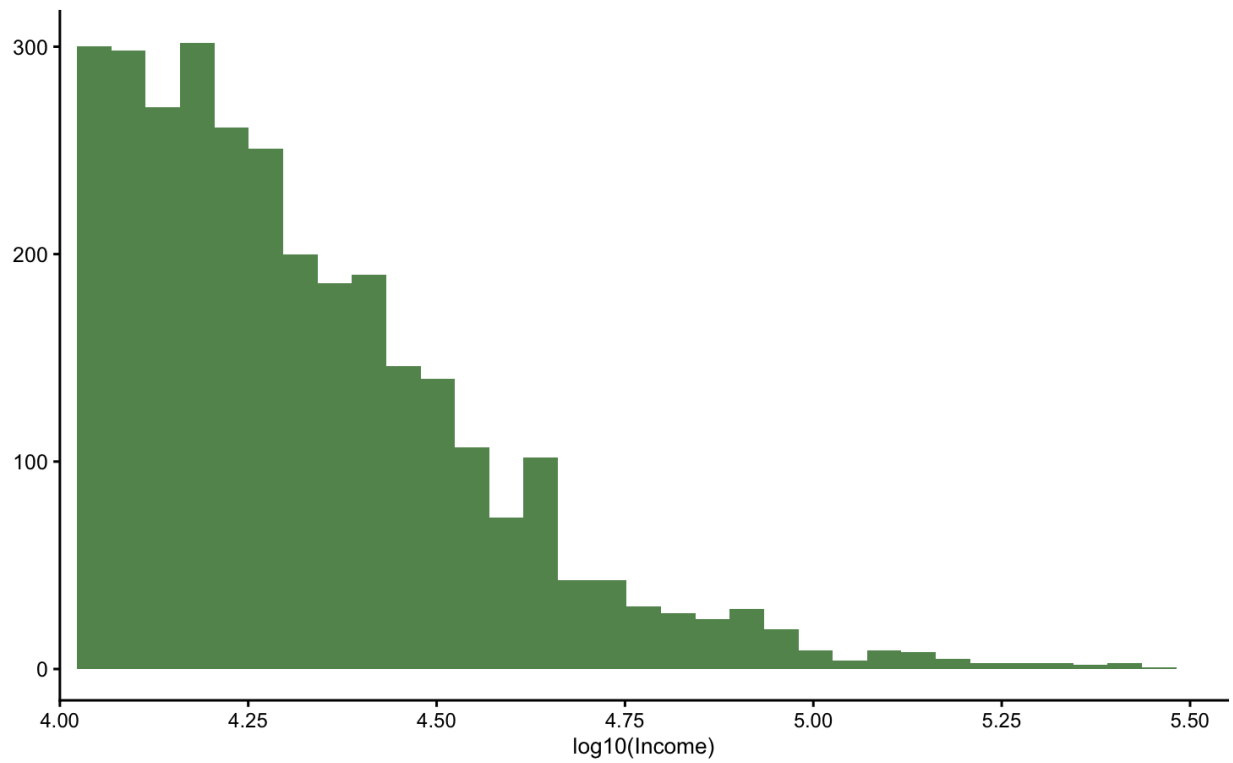
Redesign:

The fix is to apply a log10 transformation to the x axis, plotting $\log_{10}(\text{Income})$ instead of raw Income. Because a log scale converts multiplicative differences into equal additive steps, it compresses the long right tail (a handful of very high incomes) while simultaneously expanding the crowded region near zero where most ordinary households actually sit.

This works because it directly addresses the root cause of the original flaw: the data was multiplicatively distributed, and the fix simply changes the axis to a scale that matches that structure. The redesigned chart (Figure 12b) shows this clearly what was a single spike on the linear axis now spreads out into a full, examinable distribution shape with visible internal variation, while the extreme high income households now sit only a short distance away on the log scale rather than stretching the axis to a point that crushes everything else.

192524124 Problem 12 - Redesigned Log-Transformed Distribution

Log10 transformation reveals the distribution structure



Interpretation: The logarithmic scale compresses extreme high incomes and spreads the lower-income observations, making the skewed structure easier to interpret.

Figure 12b: Redesigned Log10 Transformed Distribution Structure Revealed

Problem 18 Overplotting 100,000 Point Transaction Scatterplot (5 Marks)

Scenario: A scatterplot of 100,000 transaction records is shown as a single dense black blob due to heavy overplotting, hiding any underlying pattern.

Principle Violated	Individual marks must remain visually distinguishable
Flaw Type	Overplotting mark count vastly exceeds available distinguishable pixel positions
Fix Type	2D binning (hexbin / 2D histogram) aggregate before plotting

a) Critique why this chart fails to communicate anything useful.

Critique:

This chart violates the principle that each plotted mark must remain visually distinguishable from its neighbours for a scatterplot to convey information. With 100,000 individual points drawn onto a fixed size canvas, the points in the densest region of the data overlap so heavily that they merge into one solid, undifferentiated black mass the chart has run out of distinguishable pixel positions long before it ran out of data points to plot.

A reader looking at this flawed chart can see only that 'a lot of data exists somewhere in the middle' but cannot tell whether that central region is uniformly dense throughout or actually has its own internal hotspots and variation the chart could be hiding a sharp concentration in one small area versus a broad, even spread, and there is no way to distinguish between those two very different underlying realities from the solid black blob alone. This is especially costly because the densest region is precisely where the most data and therefore the most information actually lives.

If this chart were used by a fraud detection or business analytics team to spot unusual transaction patterns, the overplotted blob would hide exactly the kind of subtle density variation (a cluster of transactions at a specific value and time combination) that such a team would be looking for, since any localized anomaly within the dense core is drawn over by thousands of ordinary points and rendered completely invisible.

192524124 Problem 18 - Flawed Scatterplot

100,000 transaction records create severe overplotting

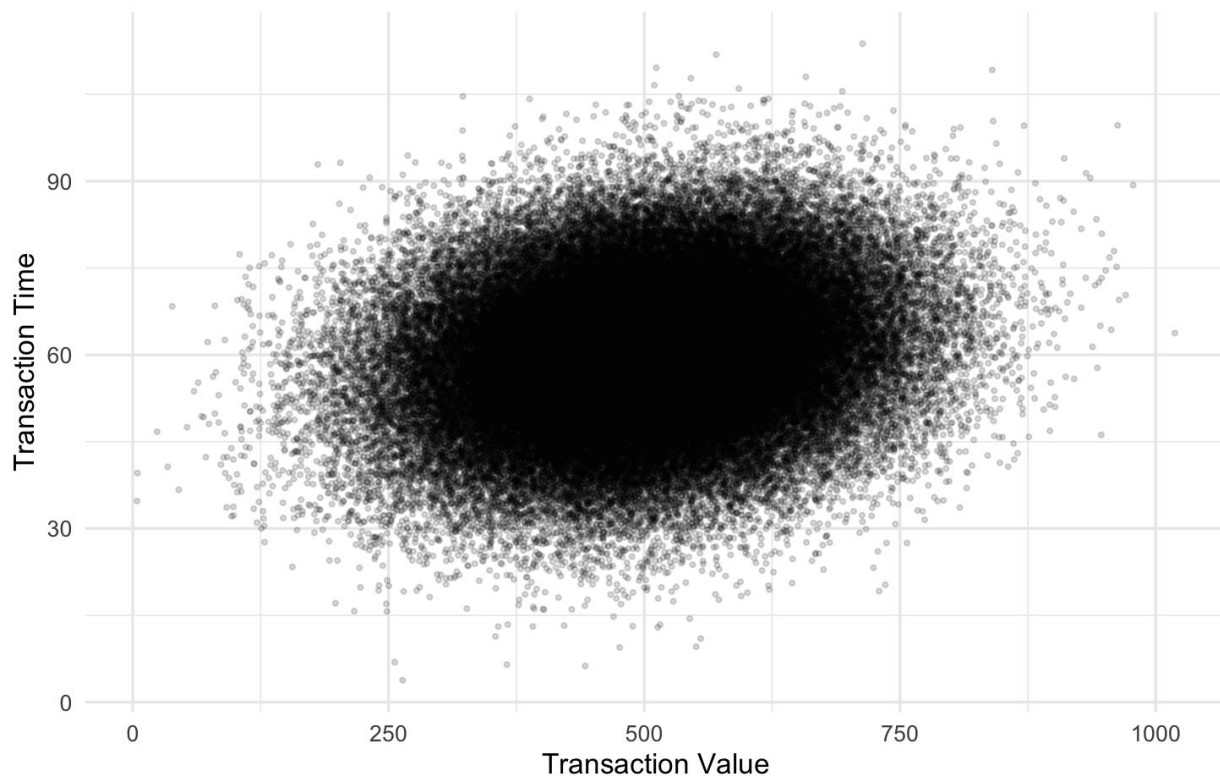


Figure 18a: Flawed Scatterplot 100,000 Points Overplotted into a Solid Blob

b) Redesign: propose an alternative chart or technique that would handle this volume of data.

Redesign:

The proposed redesign replaces the raw scatterplot with a 2D binned density chart (hexagonal or square binning): the plotting area is divided into a fixed grid of bins, the number of transactions falling into each bin is counted, and each bin is coloured according to that count using a sequential colour scale (light yellow for low counts, deep red for high counts).

This technique directly resolves the overplotting problem because the number of marks drawn is fixed by the number of bins in the grid, not by the number of underlying records so it scales to 100,000 points, or 10 million, exactly as easily as it scales to 1,000, since only the counts inside each bin change, not the number of shapes drawn. The redesigned chart (Figure 18b) reveals structure the original blob could not: a clear deep red hotspot around a transaction value of roughly 500 and a transaction time of roughly 60, showing precisely where transaction density actually peaks information that was completely invisible underneath the original chart's total visual saturation.

More broadly, overplotting is a scale of data problem rather than a chart type problem in isolation: a scatterplot is entirely appropriate for hundreds or a few thousand points, but any 'one mark per record' encoding will eventually saturate a fixed size canvas as record count grows, which is why techniques that aggregate before rendering (2D binning, density contours, or sampling a representative subset) are the standard solution once a dataset crosses roughly tens of thousands of points.

192524124 Problem 18 - Redesigned Transaction Visualization

2D binning reduces overplotting and reveals data concentration



Interpretation: Aggregating observations into bins reveals areas of high and low transaction concentration.

Figure 18b: Redesigned 2D Binned Visualization Density Hotspot Revealed

Problem 24 Deep Hierarchy Six Level Sunburst Chart (5 Marks)

Scenario: A sunburst chart is used to display 6 hierarchical levels of a company's organizational structure, resulting in tiny, unreadable outer rings.

Principle Violated	Available space must be allocated proportionally and remain legible at every hierarchy level
Flaw Type	Angular subdivision shrinks outer rings to illegible slivers as hierarchy depth increases
Fix Type	Treemap rectangular space filling subdivision

a) Critique the specific problem with using this many hierarchy levels in a sunburst chart.

Critique:

A sunburst chart divides a fixed amount of angular space among all nodes at each ring, splitting it further at every additional level of depth. With six hierarchy levels Company, Technology/Business, Software/Hardware/Sales/Finance, Development/Testing/Engineering/Support/etc., Team, and finally Unit the outrings must subdivide the same 360° among an ever growing number of nodes, so by the sixth level each Unit's wedge is both extremely thin in area and curved, forcing its label to run at an angle that is much harder to read than horizontal text.

A reader looking at this flawed chart could easily conclude that the outer ring categories (the individual Units) are unimportant or simply don't vary much in size from one another, purely because their wedges all look similarly tiny when the real explanation is a structural limitation of the sunburst chart type at this depth, not a genuine lack of variation among those Units. The chart's design flaw is being mistaken for a property of the underlying data.

If a manager used this chart to decide where to focus a restructuring effort, the illegible outer ring means they would effectively be making that decision blind to any real differences between the eight Units the very level of detail (individual teams' actual sub groups) that a restructuring decision would most need to examine is exactly the level the chart renders unreadable.

192524124 - Problem 24 - Six-Level Sunburst Chart

Flawed Visualization: Deep hierarchy creates tiny and unreadable outer rings



Interpretation: Six hierarchical levels produce very small outer segments, making labels and comparisons difficult to read.

Figure 24a: Flawed Six Level Sunburst Chart Illegible Outer Rings

b) Redesign: propose an alternative approach for visualizing deep hierarchies.

Redesign:

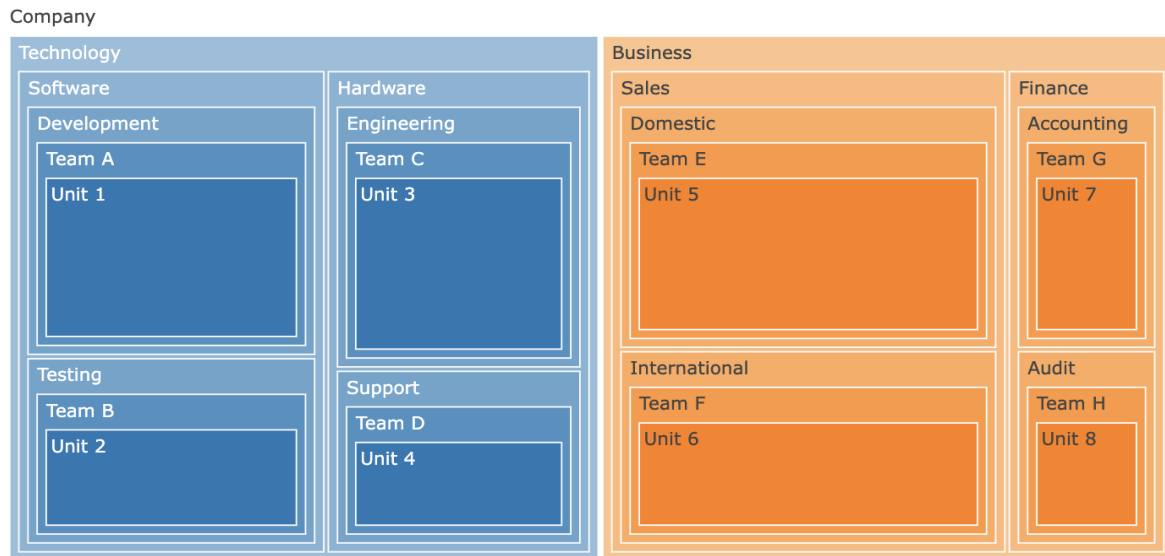
The proposed redesign replaces the sunburst with a treemap, which allocates rectangular area to each node using a space filling nested rectangle layout instead of angular wedges. Rectangles handle deep nesting far better than annular wedges because a rectangle's width and height can each shrink independently as the hierarchy nests further, rather than being squeezed into an ever thinner curved sliver, and horizontal text labels fit legibly inside a small rectangle in a way that curved, radially angled text cannot.

The redesigned treemap (Figure 24b) makes all eight leaf level Units individually visible and legibly labelled, nested inside their Team, Department, and Division containers, with area still meaningfully usable to compare sizes at any level of the hierarchy something the original sunburst's cramped outer ring could not achieve once it reached six levels of depth.

As a general guideline, sunburst charts remain readable for shallow hierarchies (roughly two to four levels) where each ring still has enough angular space per node, but for deeper hierarchies a space filling rectangular layout like a treemap or, alternatively, an indented/collapsible tree list where levels can be expanded on demand scales far more gracefully, since neither technique forces every level to compete for a shrinking share of a fixed circular boundary.

192524124 - Problem 24 - Redesigned Treemap

Redesign: Treemap improves readability of deep hierarchical structures



Interpretation: The treemap uses rectangular areas efficiently, making the six-level hierarchy easier to explore and compare.

Figure 24b: Redesigned Treemap Deep Hierarchy Remains Legible